

The perceptions towards mobile services: an empirical analysis of the role of use facilitators

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Abstract The adoption patterns for new technology such as mobile services are essential to understand for developing them further. Hence, it is important to study how willing people are to try new mobile services and do variables, which facilitate the use of the services have an effect on the consumers' perceptions of the services and their adoption readiness. This study analyses how the familiarity of mobile devices, the time the services are initially used and users' technology skills affect the initial mobile service usage experience and the likelihood of continuous usage. The results show that the duration of the use does not effect consumers' perceptions about mobile services, but familiarity of the device and user skills have an impact on the perceptions of the services.

Keywords Mobile · Adoption · Perceptions · UTAUT

1 Introduction

Mobile services adoption has become one of the most topical research areas during the past few years. There is a widespread mobile device base especially in western countries and in some Asian countries such as Korea and Japan, and the amount of new mobile applications is growing all the time. All these devices and services created are likely to promote the widespread adoption of mobile services.

At the moment, however, the popularity of mobile services is increasing quite slowly. Almost everyone has a mobile phone in his/her pocket, but the adoption of many mobile services would require the adoption of the so-called smart-phones to pick up first. Smart-phones are multi-functional mobile phones, which enable the use of various enhanced mobile services, such as MMS messaging and use of mobile internet. These phones can be loaded with information and services that help in everyday life situations.

Constant renewal of phones creates a challenge for the providers of mobile services. The understanding of how the mobile device works is a prerequisite to the use of mobile services. Many of the new features are not very easily approachable for an ordinary user as the phones are designed with more and more properties, which are increasingly complicated to use. In addition to this, the software included in the mobile phones is not necessarily easily accessible to users. To use internet-based services with mobile phones is still somewhat cumbersome. Users may also become confused with the various technologies needed to access the services, such as GPRS, WLAN, Bluetooth, infrared.

There are different mobile services offered for different use contexts. More and more mobile services are offered in situations where there is not much time available to learn to use them. For example, in a hectic use situation like a sports event, a fast learning process is essential. It is also important for the service provider who wants the system to be regarded positively after a short time of use. In this sense, it is essential to study short-term mobile services users and compare them to long-term users. Is the usage time an important aspect to consider when studying the perceptions people have

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towards mobile services after testing them? Also, based on the complicated nature of the services and hectic use contexts, how important are the users' skills in determining the perceptions towards the services?

In this paper, we attempt to answer to the above questions by focusing on studying the early stages of adoption. We analyze how willing the consumers are to become acquainted with new mobile services in a real environment. In analyzing this first "connection" to the new mobile services we also studied whether the use experience was different for consumers using their own mobile devices to those using a borrowed one. We also studied the role of individual expertise, or user skills, in this early stage of adoption.

Mobile devices are defined as the devices that are used to connect to mobile services, (wireless phones, wireless-enabled handheld computers, laptop computers, vehicle-mounted technologies, personal message pager devices) [1]. In our testing environment the mobile devices used were mobile phones and PDAs.

We define mobile services as part of mobile commerce applications, following the definition by Balasubramanian et al. [2]. They state that m-commerce applications involve communication, in which "...At least one of the parties engaged in the communication must be mobile, in the sense that his, her, or its ability to communicate is not contingent on being in a fixed physical location at a particular point in time. They classify mobile applications into eight categories based on three dimensions: the extent to which they are time critical, location sensitive and the extent to which the application is controlled (or alternatively the extent to which the use is initiated) by the user or by the provider. Services considered in this study belong to first three categories in Balasubramanian et al. taxonomy. Category 1 (time critical, location sensitive, use initiated by the user) services, such as an event calendar, a local cultural objects database and a community service, category 2 (time critical, location sensitive, initiated by the provider) services, such as mobile advertising, and category 3 (time non-critical, location sensitive, initiated by the user) services, such as various service directories.

Even though the mobile services that are considered in this study are not commercial mobile services, it can be assumed that they can be used in analyzing the consumers' needs and in studying the perceptions they have about the future mobile services. In our view the focus of mobile services adoption research should be on the idea that the future success of m-commerce depends on whether the wireless application developers are able to make good use of wireless devices and to ensure that desired applications are delivered to the

users [3]. As is shown [4], even though 97% of the respondents in their sample have owned a mobile phone for a considerable time, the use of mobile services is still in its infancy. Since mobile services have not yet spread to the masses, it is essential to let people try new kinds of mobile services and study the perceptions they have after testing them.

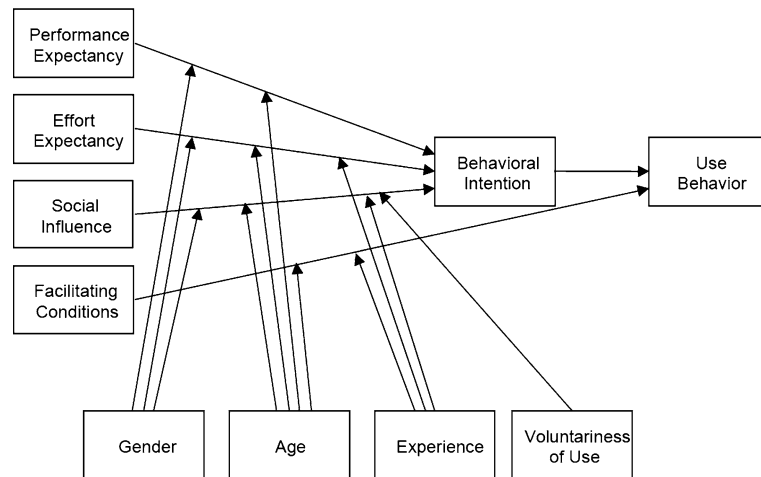
As it has been pointed out [5], the success of mobile commerce hinges on consumers' willingness to adopt this new service delivery channel and to engage in activities using systems and devices different from what they have used in the past. The knowledge of consumer behavior is therefore important in the understanding of the process of the diffusion technological innovations [6].

This paper is composed as follows: in the Sect. 2 we introduce some theoretical standpoints on consumers' adoption. In the third section the field trial is presented. Section 4 presents the analysis of the empirical data and the results. The paper ends with discussion of the results in terms of theoretical and practical implications and limitations of the study.

2 Theoretical framework for analyzing technology adoption

For the past decades the acceptance of technology has been studied quite actively, and from many alternative perspectives. These include a diffusion approach, an adoption approach and a domestication approach. Generally studies in m-commerce product and service adoption have followed the same methods [7]. In this study, the focus is in an adoption approach, because it is the most suitable theoretical background for the analysis of new service testing. It is justifiable to study foreknowledge concerning the factors affecting adoption of mobile services, even though consumers do not have much experience on these services. As is pointed out [8], adoption of end-user services in mobile commerce may be treated as technology adoption.

Various alternative approaches have used in analyzing consumers' acceptance of new technologies [9–11]. One of the more recent theories, the unified theory of acceptance and use of technology (UTAUT) (see Fig. 1.) by Venkatesh et al. [12] provides a comprehensive framework for technology adoption analysis. The model was formulated based on conceptual and empirical similarities across eight technology acceptance models. UTAUT contains four core determinants of intention and usage—performance expectancy, effort expectancy, social influence and facilitating conditions [12]. Performance

Fig. 1 UTAUT model [12]

expectancy is closely related to perceived usefulness, while effort expectancy reflects the perceived ease of use. Both usefulness and ease of use have been found as important factors in explaining technology acceptance [10]. One of the strengths of the UTAUT model is that it considers the role of several moderating variables, namely gender, age, experience and voluntariness of use [12]. These moderators are assumed to influence the significance of the four core determinants. There also exists empirical evidence indicating that these moderators actually have an effect on the significance of the core determinants. Thus, the inclusion of the moderators into the model is well justified. See e.g. Venkatesh et al. [12] for the effects of the moderators.

Although the UTAUT model is relatively new, it has inspired researchers to try its suitability in different contexts [13–16]. In this paper the scope of UTAUT is yet widened to encompass the adoption of mobile services. The focus is on the role of facilitating conditions. The effect on familiarization of mobile phones can be seen as a special character of the facilitating condition [12]. Also, the usage time and the technology use related skills could also be considered as facilitating conditions, as they both can be expected to influence the perceptions of individuals towards new technologies and technology related services. Different from the original UTAUT model, we consider the facilitating variables as moderators, which influence the other core variables in the model. This is justifiable, since in our context, the variables considered—the usage time, familiarity of the device and technology use skills—coincide more with experience (moderator in UTAUT) than with facilitating conditions (core variable in UTAUT), which in the original model is defined as “the degree to which an individual believes

that an organizational and technical infrastructure exists to support the use of the system” [12].

3 Description of the research methods

The focus of the empirical field trial was on testing m-services and technology in actual end-user environment. The field trial, SmartRotuaari2, was part of University of Oulu’s Rotuaari research project (<http://www.rotuaari.net>), and it was the fourth in a series of field tests conducted in the city of Oulu in Northern Finland. The field trial was co-ordinated from a field-office located in the pedestrian area in downtown Oulu, where the volunteering test users were able to test the services. The ordinary passers-by were targeted as the test users. If a test user did not have suitable mobile devices, the field office staff loaned such device to him/her. In many instances users were interested in testing services but had inappropriate mobile devices. The device used, own or borrowed served as the proxy for the familiarity of the device in the empirical analysis.

The services offered in SmartRotuaari2 were developed based on results from previous field trials. They included the service directories related to the city of Oulu and the downtown business association, cultural objects database, mobile ads and event calendar, which were operable with a GPRS-connection. In addition there was the flyers application, a free community service, in which the users could exchange flyers with each other over a bluetooth connection.

The users were able to use services as long as they wanted and could choose the services they were most interested in. This enabled us to study the differences in test users’ perceptions towards mobile services

Table 1 Profile of the respondents

Demographic characteristics	Number of people	Percentage	Valid percentage	Cumulative percentage
Gender				
Male	167	68.7	70.2	70.2
Female	71	29.2	29.8	100.0
Missing	5	2.1	100.0	
Total	243	100.0		
Age				
Under 18 years	16	6.6	6.8	6.8
18–24 years	53	21.8	22.4	29.1
25–34 years	100	41.2	42.2	71.3
35–49 years	48	19.8	20.3	91.6
50–64 years	20	8.2	8.4	100.0
Over 65 years				
Missing	6			
Total	243	100.0	100.0	
Occupation				
Unemployed	10	4.1	4.2	4.2
Pensioner	0.0	0.0	0.0	4.2
Student	85	35.0	35.7	39.9
Entrepreneur	21	8.6	8.8	48.7
Worker	44	18.1	18.5	67.2
Clerk	64	26.3	26.9	94.1
Executive	0.0	0.0	0.0	0.0
Other	14	5.8	5.9	100.0
Missing	5	2.1		
Total	243	100.0		
Place of residence				
Oulu region	206	84.8	86.6	86.6
Elsewhere in Finland	25	10.3	10.5	97.1
Abroad	7	2.9	2.9	100.0
Missing	5	2.1		
Total	243	100.0	100.0	

depending on the time span they had used the services. In the research setting a long-time usage meant using the services several times and for a longer than 3 h. Short-time usage meant trying the services just once or for less than 3 h. The data was collected using questionnaires. The measure of the technology related skills of the users was constructed based on the average from three separate questions inquiring the users' perceptions of his/her skill level in using mobile phones, computers and the internet. Each question was implemented on a seven-point likert-type scale, 1 = very poor,..., 7 = excellent. When the average was below or over 4, the users were categorized as unskilled or skilled, respectively.

The test users were asked questions concerning their perceptions about system's performance expectancy, effort expectancy, the effects on social influence, attitudes and the intention to use the services in the future. These core constructs and the individual items used in

the estimation of the constructs were similar to ones applied in testing the UTAUT-model [12].

The questions about the UTAUT items were implemented on seven-point likert-type scales, 1 = totally disagree,..., 7 = totally agree. The questions about the amount of use of different services were also evaluated with a seven point scale, 1 = not at all,..., 7 = very much.

4 Data analysis and results

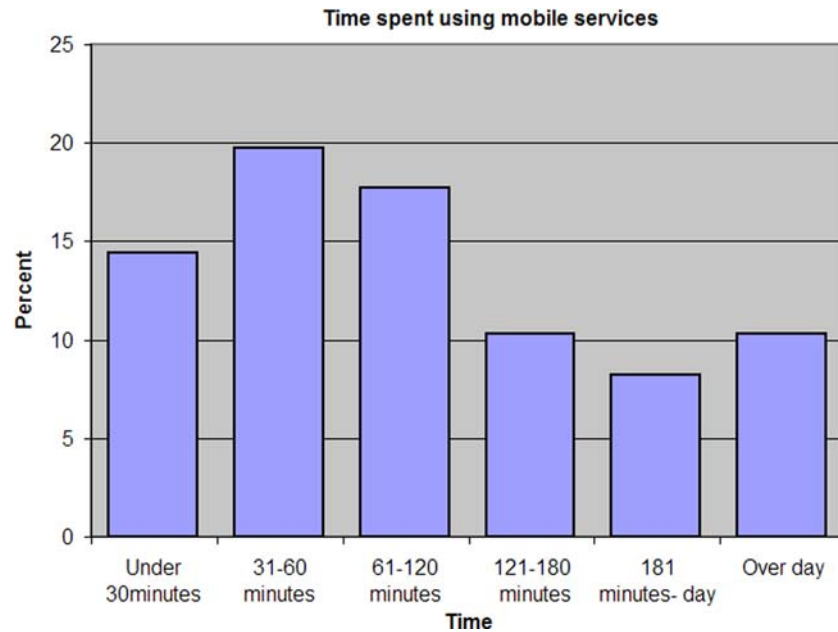
Total number of the registered users was 610 of which 39.8% filled out the questionnaire. Gender distribution of the test user group was male dominated, as over two thirds (68.7%) of the users were men. Table 1 shows that the users were relatively young, 21.8% were under 24 years and 71.3% were under 34 years old.

When examining the time used in testing the mobile services Fig. 2 shows that nearly 20% of the users used the services, according to their own estimation, 31–60 min. 17.7% of the test users used the services between 1 and 2 h. Overall, it can be said that the majority of the users used the services in a short-time or one-time nature (77% under 3 h). The users who used the services over 3 h up to several days were considered as long-time users. This division was used in the following analysis.

It was surprising that the perceptions users got from the services tested did not differ significantly depending on the time spent on using the system.

Also, when examining what kinds of services were used, the services chosen were similar except for the mobile ad service. Long-time users seem to have activated the mobile ad service and received more mobile ads (mean 4.26) than short time users (3.28) (Sig. = 0.013). The mobile ads were sent to the phones according to user's own profile settings. The number of ads was dependent on these settings and also on the customer profiling settings determined by the mobile advertisers.

It is interesting that most of our users wanted to borrow a mobile phone (71%). There may be several reasons for that. Due to the low penetration rate of smartphones suitable for using the tested services, it was hard to contact people who had compatible phones. The people who came to the field office usually did not have a suitable mobile phone. Sometimes it was also the case that people did not want to use their own GPRS -connection (due to the costs) and wanted to test the services totally free using one of the project's phones. One reason for the large amount of answers

Fig. 2 Time spent using mobile services

from the borrowed phone users was that it was relatively easy for them to fill out the questionnaire when they returned the phones to the field office, whereas the users who used their own phones could register to the services via the internet, and in many cases did not bother to fill out the questionnaire, even though it was also available online. After all, there were 610 registered users in our field trial, which refers to a lot more users with their own mobile phones than the data suggests. The data from test users indicate that 23% of users used their own mobile phones and 5.8% used PDA's. Due to the low number of the users, the PDA users were left outside the following analysis.

The perceptions of the test users based on the familiarity of the device, i.e. whether they used their own device or a borrowed one, are shown in Table 2. The results show that the users with own phones were more effected with the media, but other than that there seems to be no differences in the importance of social influence. The users with borrowed devices seem to consider the services more useful than the ones with own devices. The differences in means are statistically significant for three out of four performance expectancy items. The users with borrowed phones also seem to consider the service system more easy to use (two out of four of the items are significantly higher) and in general, they seem to have more positive attitude towards the system (all means are significantly higher). Interestingly the difference in the attitude does not indicate that they would be more likely to use the service in the future. There are no significant differences in the means of the items of the intention of future use.

Table 3 shows that there are significant differences on what and how much different mobile services were used. Mean values are relatively low on average, probably due to the relatively short use time per test user. Overall it seems that borrowers of phones did use the services more than own mobile phone users. This was quite expected, since borrowed phone users did not have to pay for any connection fees. Only the mobile ads were used more by own mobile users than by users with borrowed phones. The greatest difference on using service usage can be seen to be the flyer service (means 3.88/1.98). This was an expected result, since it was the only service that had to be downloaded to the users' phones at the field trial office. Generally, the results on services usage show that the event calendar and mobile ads were the most used services, although the differences between the popularity of individual services are relatively small.

When we consider user technology skills as the moderating variable, we see that there are some notable differences between skilled and unskilled users. The results presented in Table 4 show that performance expectancy, or the usefulness of the service, was regarded somewhat higher by the skilled users. The means of two out of four performance expectancy items have a statistically significant difference. Skilled users also seem to consider the services as more easy to use. All effort expectancy items show a statistically significant difference in the means.

Contrary to the results of prior studies such as Venkatesh et al. [12], the social influence seems to be stronger for the skilled users than for the unskilled ones (significant difference in 3 out of 4 means).

Table 2 Perceptions of the test users and the familiarity of the device

Variables and items	Mobile device	Mean	SD	Sig. (two-tailed)
Performance expectancy				
In my opinion, the system is useful	Borrowed	4.98	1.37	0.106
	Own	4.60	1.42	
Using the system was beneficial to me	Borrowed	4.11	1.52	0.006
	Own	3.38	1.72	
The system makes it more interesting to visit the town center	Borrowed	4.60	1.60	0.047
	Own	4.04	1.80	
The system was better than the traditional sources of similar information	Borrowed	3.92	1.57	0.023
	Own	3.30	1.47	
Effort expectancy				
Using the system was clear	Borrowed	4.00	1.50	0.458
	Own	3.78	1.81	
The system was easy to use	Borrowed	4.04	1.63	0.000
	Own	5.53	1.82	
Establishing a connection was effortless	Borrowed	4.11	1.79	0.342
	Own	3.82	1.92	
Viewing the content was easy	Borrowed	4.58	1.56	0.053
	Own	4.05	1.85	
Social influence				
Hearing about the system in the media influenced me to take it in use	Borrowed	3.34	2.13	0.028
	Own	4.14	2.06	
People who are important to me think that I should use the system	Borrowed	3.83	1.99	0.528
	Own	3.58	2.21	
The people close to me think that I should use the system	Borrowed	3.63	2.04	0.837
	Own	3.55	1.80	
Seeing other people using the system affected my use	Borrowed	2.48	1.86	0.118
	Own	2.05	1.51	
Attitude				
I liked using the system	Borrowed	4.61	1.46	0.006
	Own	3.91	1.58	
Using the system was enjoyable	Borrowed	4.07	1.64	0.056
	Own	3.55	1.53	
Using the system was fun	Borrowed	5.15	1.54	0.002
	Own	4.30	1.65	
Intention to use				
I am going to use a similar system also in the future	Borrowed	4.77	1.51	0.312
	Own	5.05	1.76	
If available, I would use a similar system in the future	Borrowed	5.20	1.54	0.345
	Own	5.44	1.53	
I intend to use the system in the near future	Borrowed	4.54	1.81	0.303
	Own	4.86	1.59	

t test for equality of means, H_0 means are equal

Perhaps the people who consider themselves as skilled technology users also regard themselves as pioneers and consider it important to be among the first users of any new technologies. Therefore, they are more easily influenced by media and/or recommendations of others, especially when the informants belong to the same social reference group.

The attitudes of skilled users are more positive towards the services. In this case, the difference in the attitudes is also reflected on the intention to use the system again.

The results of the usage of the services of skilled and unskilled users are shown in Table 5. There are no

statistically significant differences in the popularity of the services, although the difference of the means for the mobile ads and city of Oulu's service directory are almost statistically significant. In both cases the services seem to be more popular among the skilled users. In general, the event calendar seems to be the most popular service.

5 Conclusions

The main objective of this study was to examine how different variables that facilitate the use of mobile

Table 3 Usage of the services by the familiarity of the device

Services used	Device	Mean	SD	Sig (two-tailed)
Event calendar	Own	3.19	2.06	0.000
	Borrowed	4.43	1.96	
Mobile ads	Own	4.39	2.29	0.004
	Borrowed	3.33	2.15	
Service directory, SmartRotuaari2	Own	2.51	1.78	0.001
	Borrowed	3.68	2.10	
Flyers	Own	1.98	1.99	0.000
	Borrowed	3.88	2.23	
Cultural objects database	Own	2.24	1.74	0.001
	Borrowed	3.38	2.20	
Service directory, city of Oulu	Own	2.56	1.78	0.121
	Borrowed	3.09	2.06	
Service directory, business association	Own	2.24	1.68	0.003
	Borrowed	3.19	2.10	

t test for equality of means, H_0 means are equal

services, the time of usage, familiarity of the mobile device and users' technology skills, effect users' perceptions of performance expectancy (usefulness) of the services, effort expectancy (ease of use), social influence, attitude towards the services and the intention to use the services again.

The results showed that the perceptions users got from the services tested did not differ a lot depending on the time spent on using them. This is an interesting finding as the general hypothesis is that people get deeper into new service or technology if they have more time to get used to it. This result encourages the mobile service developers to develop more intuitive services since the first impression seems to be the one that lasts.

Even though the time spent did not have an effect on perception, the familiarity of the device used did have an impact on users' perceptions of the service. Borrowers considered the system being more useful and easier to use than the users of own phones and they also had more positive attitudes towards the system. One explanation for this might be that the small imperfections of the system were more tolerable while using a loaned device and the possible problems would not concern the user. Also, the fact that a borrowed device is not as personal as own mobile device may increase the tolerance. One notable result was that the difference in the attitudes was not reflected in the intention to use the system again.

Users' technology skills also affected their perceptions of the services. Skilled users considered the system more useful and easier to use. An interesting finding was that the influence of hearing about the system in the media and the recommendations of others were also more significant for skilled users.

There also seems to be a connection between the attitude towards the services and the intention to use them again, since they both were considered significantly higher by the skilled users.

Based on the results the use facilitators had only minor effects on the popularity of different services. The result indicating that the use with borrowed phones was more active was most probably due to the fact that users had no connection fees when using borrowed phones. All in all the most popular service seemed to be the event calendar.

Our results give rise to some important implications. Firstly, the fact that increased user skills generate more positive perceptions towards mobile services and increase the likelihood of continuous service use stresses the importance of enhancing the technology skills of general public and potential mobile service users. This can be done by e.g. organizing mobile service/mobile phone use tutoring workshops, including easily accessible user aids (manuals, navigational user guidance, etc.) or providing comprehensive hands-on user guidance in the situations where new phones are purchased. The increase in people's perceptions about their technology use skills will increase the adoption of new mobile services. Secondly, since the tolerance of service imperfections seems to decrease with the familiarity of the device, it is essential that services, which are introduced to general public, must be carefully designed and implemented. Long-term success of mobile services requires that all pieces of the puzzle—devices, connections, service interfaces and content—must be in place.

From a theoretical standpoint, the results presented in this paper contributed to the existing literature in a number of ways. First, the paper makes a contribution

Table 4 Perceptions of the test users and user technology skills

Variables and items	Mobile device skills	Mean	SD	Sig. (two-tailed)
Performance expectancy				
In my opinion, the system is useful	Unskilled	4.73	1.27	0.159
	Skilled	5.01	1.41	
Using the system was beneficial to me	Unskilled	3.62	1.54	0.014
	Skilled	4.20	1.66	
The system makes it more interesting to visit the town center	Unskilled	4.15	1.58	0.067
	Skilled	4.59	1.71	
The system was better than the traditional sources of similar information	Unskilled	3.55	1.48	0.113
	Skilled	3.94	1.67	
Effort expectancy				
Using the system was clear	Unskilled	3.57	1.36	0.000
	Skilled	4.33	1.67	
The system was easy to use	Unskilled	3.52	1.64	0.000
	Skilled	5.03	1.64	
Establishing a connection was effortless	Unskilled	3.60	1.63	0.008
	Skilled	4.28	1.98	
Viewing the content was easy	Unskilled	4.09	1.62	0.005
	Skilled	4.77	1.70	
Social influence				
Hearing about the system in the media influenced me to take it in use	Unskilled	2.74	1.86	0.000
	Skilled	3.98	2.27	
People who are important to me think that I should use the system	Unskilled	3.27	1.99	0.015
	Skilled	4.08	2.05	
The people close to me think that I should use the system	Unskilled	3.05	1.86	0.010
	Skilled	3.91	2.07	
Seeing other people using the system affected my use	Unskilled	2.54	1.88	0.251
	Skilled	2.24	1.74	
Attitude				
I liked using the system	Unskilled	4.23	1.43	0.174
	Skilled	4.53	1.61	
Using the system was enjoyable	Unskilled	3.56	1.48	0.004
	Skilled	4.22	1.66	
Using the system was fun	Unskilled	4.47	1.65	0.006
	Skilled	5.12	1.58	
Intention to use				
I am going to use a similar system also in the future	Unskilled	4.14	1.62	0.000
	Skilled	5.19	1.47	
If available, I would use a similar system in the future	Unskilled	4.76	1.57	0.001
	Skilled	5.48	1.49	
I intend to use the system in the near future	Unskilled	3.94	1.76	0.000
	Skilled	4.94	1.70	

t test for equality of means, H_0 means are equal

to mobile service literature by providing insights on the necessity of studying the needs and perceptions of end-users. It also contributed to the discussion of adoption of new technology: a mobile service. UTAUT model was used in finding out the perceptions people got from the mobile service. Especially the facilitating conditions were concentrated on, which were effected by time and device used. Also the intention of using the mobile service again was taken into consideration in this study.

However, the work is still at an early stage and certain limitations concerning the research setting should be noted in order to guide future research. For

example, more research is needed to leverage the findings and provide better and more in-depth implications for both theory and practice. The second limitation concerns the sample. Although the sample size was quite large, it consisted of Finnish consumers only. This has an effect on the generalization of the findings. The third limitation may be that there were only a fixed number of services of which the consumers could choose from and evaluate. If there would be more possibilities for the consumers to choose from, the results might be different. Despite these limitations, the results gave rise to several important implications. An interesting avenue for further research would be to

Table 5 Usage of the services by user technology skills

Services used	Mobile device skills	Mean	SD	Sig (two-tailed)
Event calendar	Unskilled	4.13	2.00	0.532
	Skilled	4.31	2.05	
mobile ads	Unskilled	3.14	2.13	0.067
	Skilled	3.75	2.23	
Service directory, SmartRotuaari2	Unskilled	3.37	1.99	0.314
	Skilled	3.70	2.12	
Flyers	Unskilled	3.16	2.22	0.647
	Skilled	3.32	2.38	
Cultural objects database	Unskilled	3.26	2.20	0.622
	Skilled	3.09	2.12	
Service directory, city of Oulu	Unskilled	2.75	1.94	0.069
	Skilled	3.34	2.06	
Service directory, business association	Unskilled	2.91	1.97	0.416
	Skilled	3.18	2.13	

t test for equality of means, H_0 means are equal

increase the generalizability of our results and implications by conducting similar studies across countries and with as many different types of mobile services.

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References

1. Tarasewich P, Nickerson RC, Warkenti M (2002) Issues in mobile e-commerce. *Commun Assoc Inform Sys* 8:41–64
2. Balasubramanian S, Peterson RA, Jarvenpaa SL (2002) exploring the implications of m-commerce for markets and marketing. *Acad Mark Sci* 30(4):348–362
3. Shim JP, Shim JM (2003) M-commerce around the world: mobile services and applications in Japan, Korea, Hong Kong, Finland and the US. *Decision Line* September/October
4. Aarnio J, Enkenberg A, Heikkilä J, Hirvola S (2002) Adoption and use of mobile services empirical evidence from a finnish survey. *Proceedings of the 35th annual Hawaii international conference on system sciences*. Hawaii
5. Bruner II GC, Kumar A (2005) Explaining consumer acceptance of handheld Internet devices. *J Bus Res* 58:553–558
6. Green N, Harper RHR, Murtagh G, Cooper G (2001) Configuring the mobile user. *Sociological and Industry Views. Personal Ubiquitous Computing* 5:146–156
7. Han S (2002) A framework for understanding adoption and diffusion processes for m-commerce products and services and its potential implications for planning industry foresight. Conference paper IRIS25 August, Falkenberg, Sweden
8. Pedersen PE, Methlie LB, Thorbjørnsen H (2002) Understanding mobile commerce end-user adoption: a triangulation perspective and suggestions for an exploratory service evaluation framework. *Proceedings of the 35th Hawaii international conference on system sciences*, Hawaii
9. Ajzen I, Fishbein M (1977) Attitude–behavior relations: a theoretical analysis and review of empirical research. *Psychol Bull* 84:5
10. Davis FD (1989) Perceived usefulness, perceived easy of use and User acceptance of information technology. *MIS Q* 319–340
11. Rogers EM (1995) Diffusion of innovations. The Free Press, New York
12. Venkatesh W, Morris MG, Davis GB, Davis FD (2003) User acceptance of information technology: toward a unified view. *MIS Q* 27(3):425–478
13. Rosen PA (2004) The effect of personal innovativeness in the domain of information technology on the acceptance and use of technology. A working paper 2004
14. Schaper L, Perven G (2004) A model of information and communication technology acceptance and utilisation by occupational therapist. *Decision support in an uncertain and complex world: the IFIP TC8/WG8.3 international conference*
15. Mallat N (2004) Theoretical constructs of mobile payment adoption. Conference paper, IRIS27 August 14–17, Falkenberg
16. Anderson JE, Schwger PH (2004) SME Adoption of wireless LAN technology: applying the UTAUT model. *Proceedings of the 7th annual conference of the southern association for information systems*